

- *Project Title: Milk Collection and Quality Tracking system - Gujarat.*
- *Domain: AG-01 (Agri - Tech)*
- *Team Name: Code Crafters*
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- *Event: Udbhava X'26, Coding Club SR University*



Problem Scenario

The Real-World Situation:

In rural villages, farmers bring milk twice daily to collection centers where staff measure quantity and test quality. Due to large queues and time pressure, data is quickly written in notebooks. Farmers usually leave without clear confirmation of their records or payments.

The Current Failure: The manual system leads to frequent errors in data entry, loss or damage of records, and lack of transparency. Farmers are unaware of how their payments are calculated, and there is no centralized system to manage data efficiently. This results in confusion, inefficiency, and mistrust.



Current Pain Points (Why it Breaks)

- The existing system has several critical pain points:
- Manual data entry leads to frequent errors
- Records can be lost, damaged, or incomplete
- No real-time updates for farmers
- Lack of transparency in quality and payment calculation
- Manual payment calculation is time-consuming and inaccurate
- No centralized system across multiple collection centers
- Difficulty in accessing historical data and generating reports

Why the System Breaks

- The system fails mainly because it depends entirely on manual processes.
- Human errors increase under time pressure
- Paper-based records have no backup
- No automation leads to delays and inefficiency
- Data is scattered across locations with no integration
- Lack of transparency creates mistrust between farmers and cooperatives



Objectives & Success Criteria

PRIMARY OBJECTIVE

The objective of this system is to digitize the milk collection process and eliminate manual record keeping. It aims to ensure accurate tracking of milk quantity and quality for each farmer. The system will automate payment calculations to reduce errors and delays. It also focuses on improving transparency so farmers can easily access their data. Additionally, it provides centralized data management for efficient monitoring across multiple collection centers.

SUCCESS CRITERIA

The system is successful if it accurately records milk quantity and quality without errors.

Payment calculations should be automated and transparent for farmers.

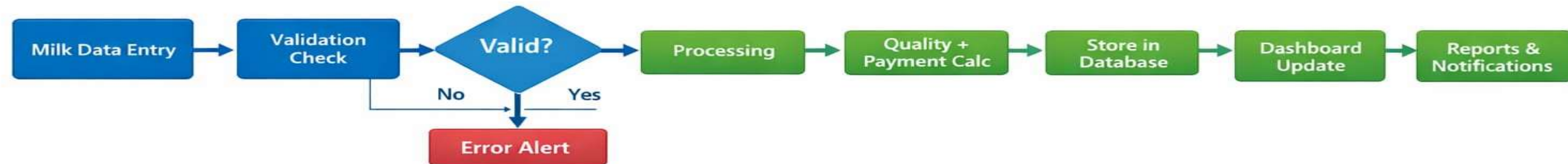
Data must be securely stored with easy access and no loss.

Reports should be generated quickly for better decision-making.

The system should handle multiple centers efficiently.



Proposed Solution Strategy



PROPOSED STRATEGIES:

- **Validation:** Check data instantly to avoid errors
- **Digital Data Entry:** Replace notebooks with system-based entry
- **Automation:** Automatically calculate quality and payment
- **Central Storage:** Store all data securely in one database
- **Transparency:** Farmers and admins can view data anytime
- **Reports:** Generate daily and payment reports
- **Scalability:** Support multiple centers and many farmers



Architectural Overview (3-Tier)

Frontend Layer (React)

Provides user-friendly dashboards for Staff/Admin and Farmers. Handles data input, display, and communicates with backend using REST APIs.

Backend Logic Layer

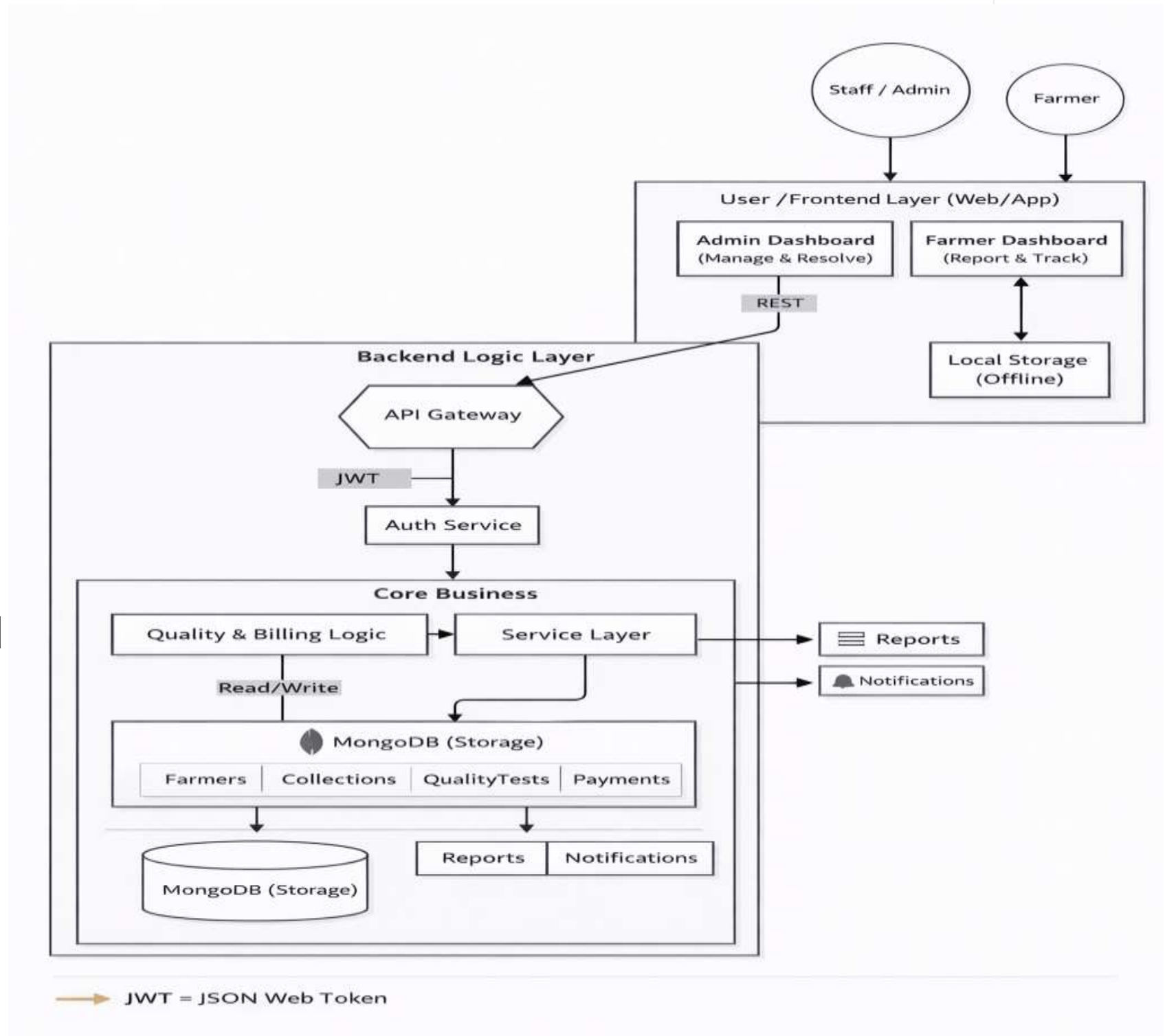
Manages authentication (JWT), data validation, and core business logic.

Processes milk collection, quality analysis, and payment calculation using service modules.

Data Persistence Layer (MongoDB)

Stores structured data such as farmers, collections, quality tests, and payments.

Ensures secure storage, quick retrieval, and historical tracking of records.



Data Flow / Workflow

Submission

Staff selects the farmer and enters milk details such as quantity and quality. then submits the data.

Validation

The system checks whether the entered values are within valid limits.

If invalid → shows error and stops

If valid → proceeds further

Processing

The system calculates milk quality and automatically computes the payment amount.

Data Storage

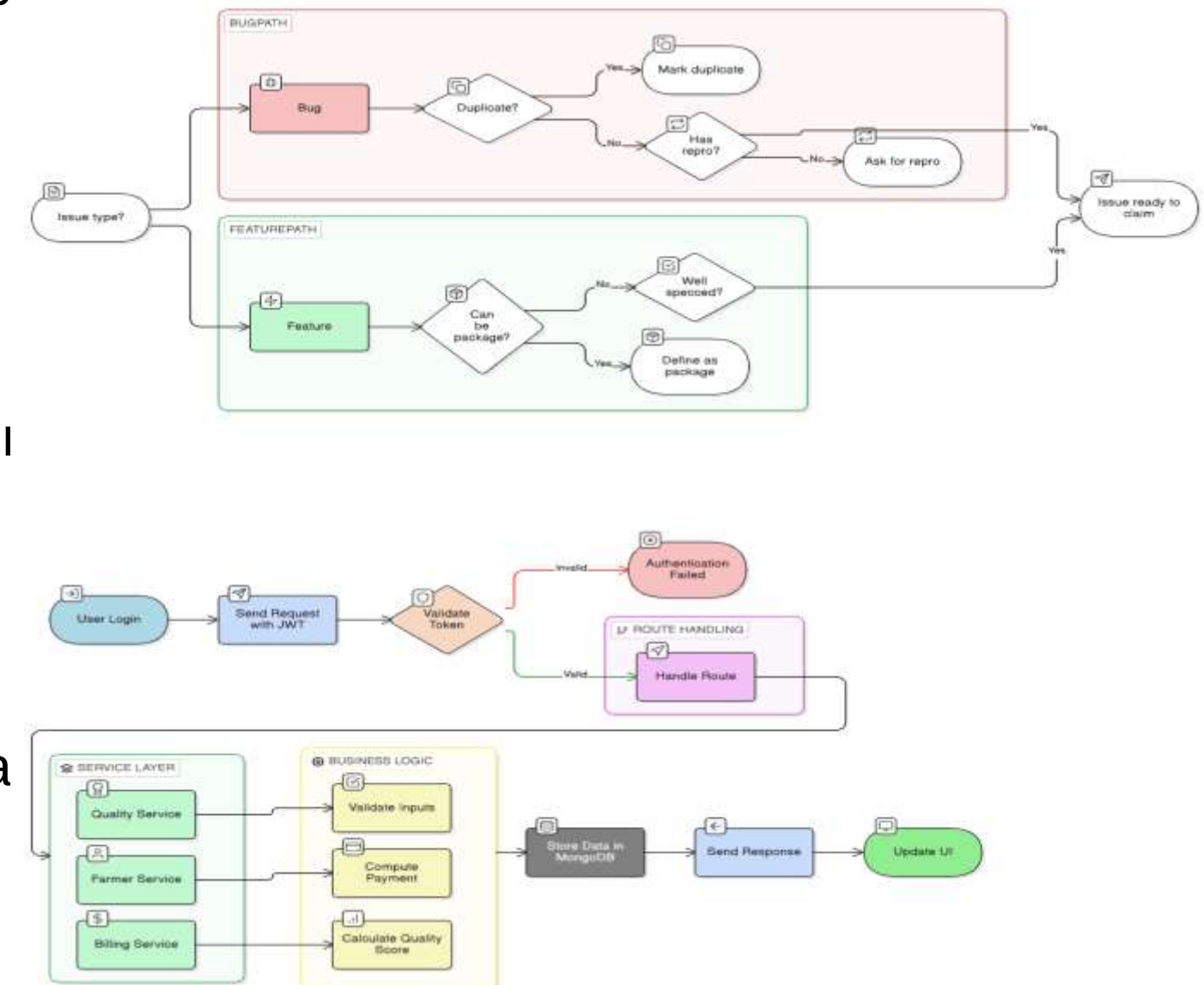
All details are stored securely in the database for future reference.

Visualization

The dashboard updates instantly, showing collection data, quality status, a payment details.

Update / Resolution

Admin or staff can update records or payment status, and changes are reflected immediately in the system.



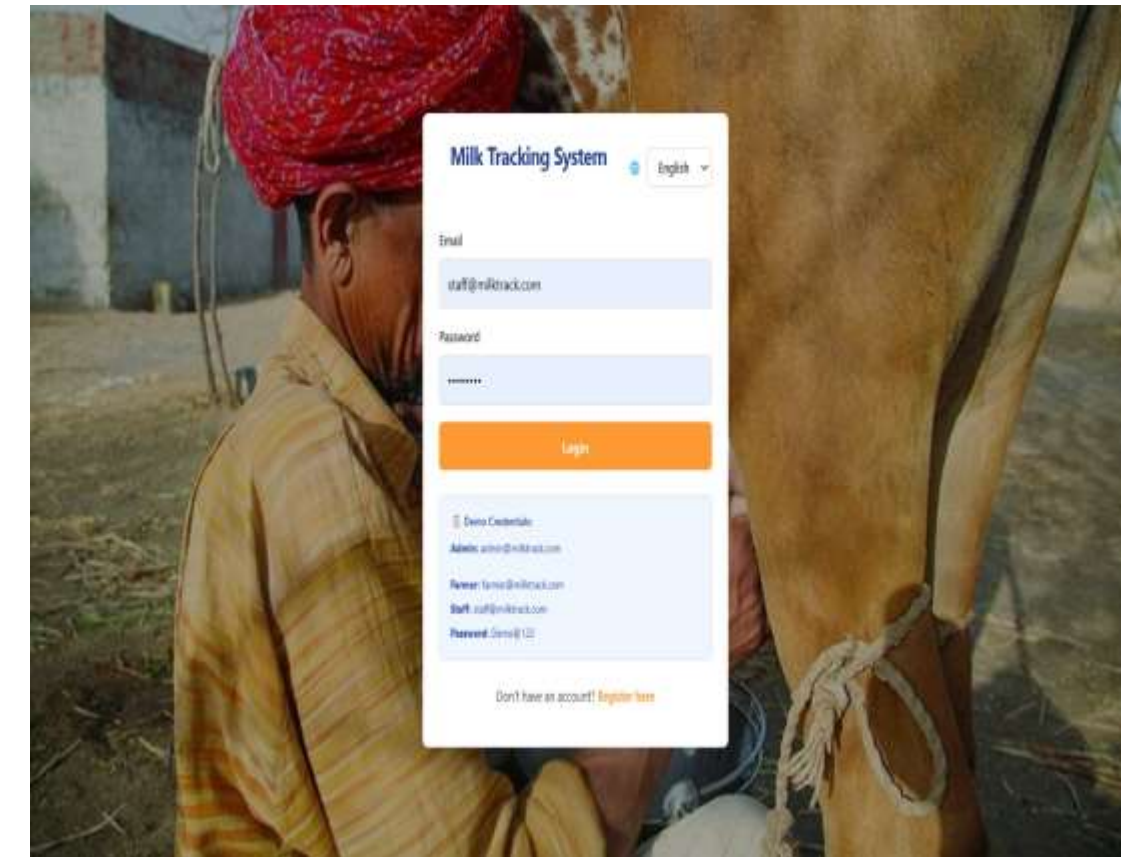
Implementation features

- Role-based Login (Farmer/Staff/Admin)
- Farmer Registration & Profile Management
- Daily Milk Quantity & Quality Entry Forms with Dropdowns
- Automated Payment Calculation & Transaction Records
- Admin Dashboard with Collection & Payment Statistics Cards
- Historical Data Tracking & Search Functionality
- Farmer Notifications for Payments & Quality Updates
- Visual Alerts for High-Quality / Low-Quality Milk Entries

Link Requirements:

GIT HUB DEMO: [budagamhaasini16/Code-Crafters---Milk-Tracking-System](https://github.com/budagamhaasini16/Code-Crafters---Milk-Tracking-System): A system designed to digitize milk collection by recording quantity, analyzing quality, and automating payment calculations with improved accuracy and transparency.

LIVE DEMO: <https://code-crafters---milk-tracking-system-go6k77jrmro7y6xftyakmz.streamlit.app/>



Impact & Benefits

Farmers

- Clear payment transparency
- No confusion or disputes
- Easy access to records

Staff

- Faster data entry
- Reduced manual errors
- Easy management of daily records

Cooperative/Admin

- Centralized data management
- Better monitoring & reporting
- Improved efficiency across centers



Internet Dependency

The system requires stable internet connectivity for data entry and access. In rural areas, poor network availability may affect real-time usage and system reliability.

Limited Real-World Testing

The current system is developed as a prototype and has not been extensively tested in real village environments, which may reveal additional practical challenges.

Lack of Hardware Integration

The system does not currently integrate with automatic milk testing devices, so quality data still depends on manual input, which may affect accuracy.



Future Scope & Roadmap

Phase 2

Mobile application for farmers

Online payment integration (UPI/Bank)

Multi-language support

Phase 3

IoT-based automatic quality testing

AI-based prediction & fraud detection

Cloud deployment for large-scale usage

